EXP NO: 9

DETERMINATION OF Z PARAMETES OF TWO PORT NETWORK.

AIM:

To calculate and verify Z parameters of a two port network using multisim simulation

APPARATUS REQUIRED:

S.NO	Components	Specification	Quantity
1.	power supply		1
2.	multimeter	—	1
3.	Resistors	10KΩ, 10KΩ 4.7KΩ	
4.	Bread board	—	1
5.	connecting wires	—	—
6.	pc with multisim simulation	—	—

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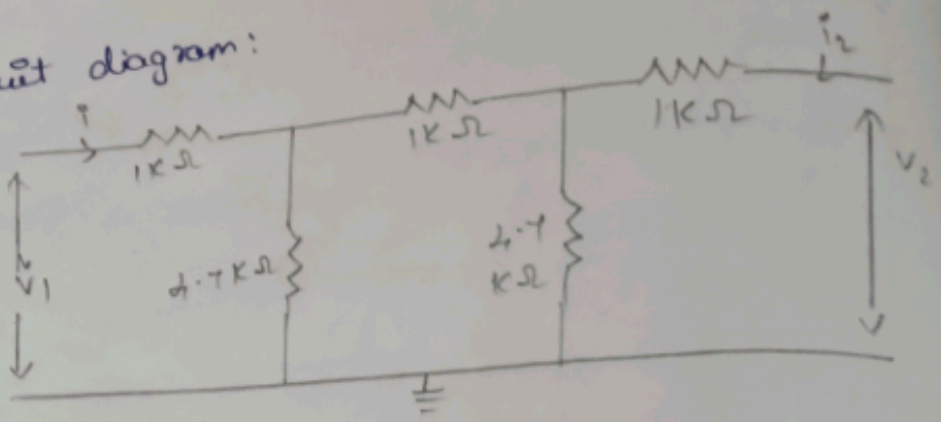
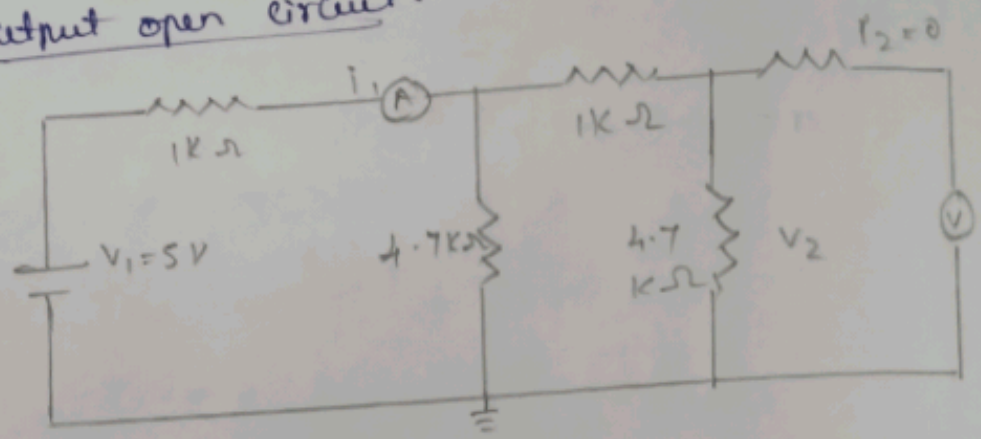
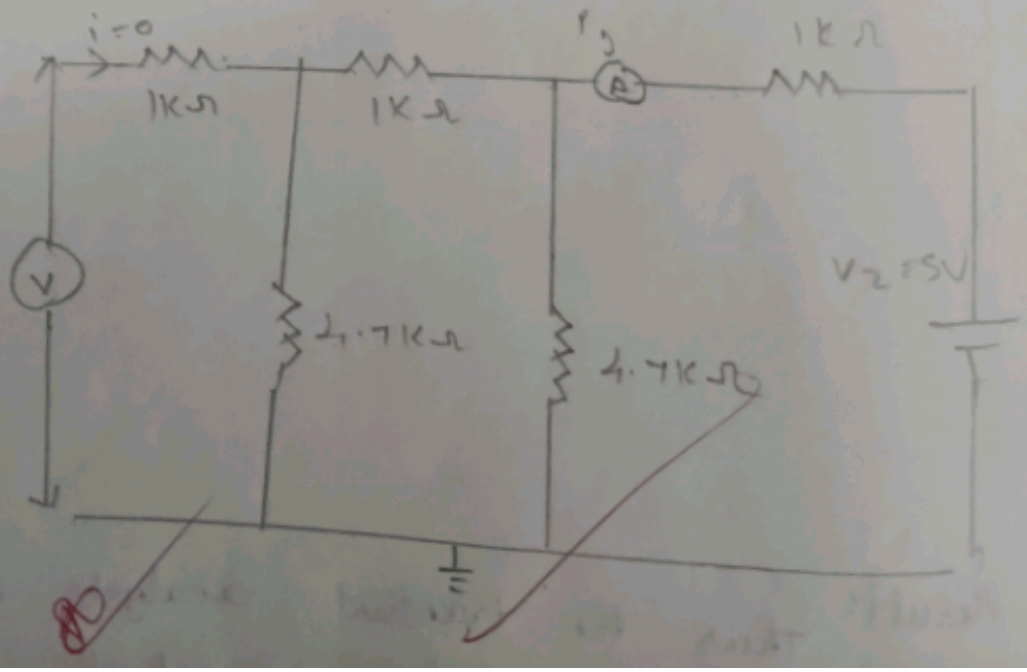
Circuit diagram:



Output open circuit:



Input open circuited:



procedure:

a) Connect the circuit as shown in fig and switch 'on' the experimental board

b) first open the o/p terminal and supply 5v to I/p terminal. Measure o/p voltage and I/p current.

c) secondly, open I/P terminal and supply 5v to o/p terminal. Measure I/P voltage and o/p current using multi-meter.

d) Calculate the values of z parameters using equation 1 and 2

e) Switch off the supply after taking the readings.

observation:

S.NO	output open circuited ($I_2=0$)			$Z_{21} = \frac{V_2}{I_1}$ (Ω)	$Z_{11} = \frac{V_1}{I_1}$
	V_1	V_2	I_1		
1,	5	2	2	$Z_{21} = \frac{2}{2mA}$ $= 1K\Omega$	$Z_{11} = \frac{5}{2mA}$ $= 2.5K\Omega$
2,	6	3	3	$Z_{21} = 3/3mA$ $= 1K\Omega$	$Z_{11} = 6/3$ $= 3K\Omega$

S.NO	Input open circuited ($I_1=0$)			$Z_{12} = \frac{V_1}{I_2}$	$Z_{22} = \frac{V_2}{I_2}$
	V_2	V_1	I_2		
1,	5	2	2	$Z_{12} = 5/2$ $= 2.5K\Omega$	$Z_{22} = 5/2$ $= 2.5K\Omega$
2,	6	3	3	$Z_{12} = 3/3$ $= 1K\Omega$	$Z_{22} = 6/3$ $= 2K\Omega$

All verified

Theoretical verification:

$$Z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0}$$

$$= \frac{5}{2mA} = 2.5K\Omega$$

$$Z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0}$$

$$= \frac{2}{2mA} = 1K\Omega$$

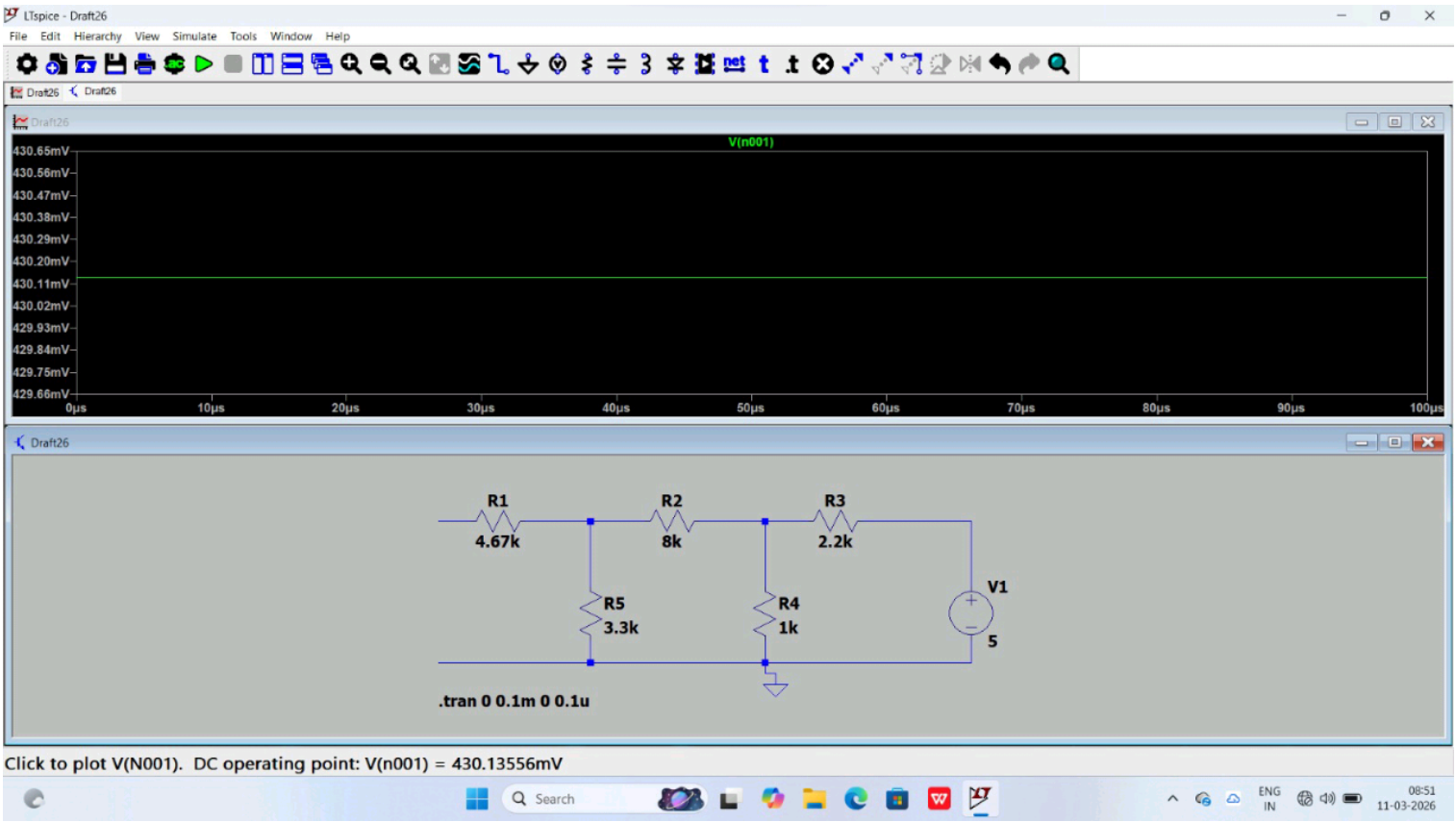
$$Z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0} = \frac{2}{2mA} = 1K\Omega$$

$$Z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0} = \frac{5}{2mA} = 2.5K\Omega$$

$$Z = \begin{bmatrix} 2.5K\Omega & 1K\Omega \\ 1K\Omega & 2.5K\Omega \end{bmatrix}$$

Marks split-up:

S.NO	Description	Marks allotted	Marks obtained
1.	Experimental setup / Connections	20 marks	20
2.	observation / Readings	20 marks	20
3.	Calculations / Result	20 marks	20
4.	viva	20 marks	20
5.	Record	20 marks	20
		Total :	100


LTspice - Draft26
File Edit Hierarchy View Simulate Tools Window Help
Draft26
V(n001)
430.65mV
430.56mV
430.47mV
430.38mV
430.29mV
430.20mV
430.11mV
430.02mV
429.93mV
429.84mV
429.75mV
429.66mV
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
R1
4.67k
R2
8k
R3
2.2k
R5
3.3k
R4
1k
V1
5
.tran 0 0.1m 0 0.1u
Click to plot V(N001). DC operating point: V(n001) = 430.13556mV
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Result:

The z-parameters of the two port network has been calculated and verified by using multisim simulation.